

Explainable Deep Learning Framework for Pneumonia Detection and Classification Using Transfer Learning and Grad-CAM

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Abstract—Pneumonia remains a major respiratory disease requiring timely and accurate diagnosis, particularly in clinical settings where radiological expertise and diagnostic resources may be limited. Chest X-ray imaging is widely used as a first-line diagnostic modality due to its accessibility, low cost, and rapid acquisition. However, reliable interpretation of chest X-ray images is challenging because visual manifestations of COVID-19, bacterial pneumonia, viral pneumonia, and normal lung conditions may overlap. Recent advances in deep learning have demonstrated strong potential for automated chest disease detection, but many existing studies focus mainly on binary classification and provide limited clinical interpretability. This paper proposes an explainable deep learning framework for four-class chest X-ray classification involving Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia cases. The proposed study evaluates three modern convolutional architectures: DenseNet121, EfficientNetV2B3, and EfficientNetV2S. Transfer learning is employed to improve model convergence and performance under medical image data constraints. Grad-CAM is integrated to provide visual explanations by highlighting image regions that influence model predictions. In addition to classification, the proposed system includes web-based and mobile-friendly deployment, historical prediction management, and automated PDF medical report generation. Experimental results show that DenseNet121 achieved the highest accuracy of 85.31% (95% CI 83.3–87.1%), followed by EfficientNetV2B3 with 84.30% and EfficientNetV2S with 80.54%. COVID-19 and Normal cases were recognized with recall of at least 0.97, whereas 76% of DenseNet121 errors were confusions between bacterial and viral pneumonia. Lung-restricted Grad-CAM maps are reported together with the share of attention inside the lung fields, and the results are compared with previously published methods. The framework is presented as a decision-support prototype that requires external clinical validation before use.

Keywords—Clinical Decision Support, COVID-19, Deep Learning, Grad-CAM, Pneumonia Detection

I. INTRODUCTION

Pneumonia is one of the leading causes of morbidity and mortality worldwide, responsible for millions of deaths annually and disproportionately affecting children under five years of age and elderly populations [1]. It is caused by a variety of pathogens including bacteria, viruses, fungi, and atypical organisms, each presenting with distinct radiographic and clinical characteristics that require different treatment approaches. The timely and accurate differentiation of pneumonia subtypes including bacterial pneumonia, viral pneumonia, and COVID-19 from normal lung conditions is therefore a clinical necessity that directly influences patient

management, antibiotic prescription decisions, and isolation protocols. Chest X-ray imaging remains the most widely adopted first-line diagnostic modality for pulmonary diseases due to its accessibility, low cost, rapid acquisition, and broad availability across clinical settings, including low-resource environments. However, reliable interpretation of chest X-ray images presents substantial challenges even for experienced radiologists. Visual manifestations of COVID-19, bacterial pneumonia, viral pneumonia, and normal lung conditions frequently overlap, and radiographic findings can be subtle or non-specific, particularly in early-stage infections. Radiologist availability is uneven across institutions and geographic regions, and high patient volumes in emergency and intensive care settings can introduce diagnostic delays that adversely impact patient outcomes. Automated and explainable computational tools therefore offer critical support for reducing radiologist workload, improving diagnostic consistency, and providing rapid preliminary assessments in high-volume clinical environments.

Deep learning has emerged as the dominant paradigm in medical image analysis, as convolutional neural networks (CNNs) automatically extract hierarchical visual representations from low-level edges and textures to high-level structural patterns without the need for handcrafted feature engineering [2]. Foundational research established that deep networks achieve strong image recognition performance given appropriate architectures and training strategies [2], motivating the development of AI systems capable of supporting clinical authority while reducing diagnostic burden [1], [3]. Large benchmark datasets have greatly accelerated chest X-ray research. The ChestX-ray8 dataset established a hospital-scale multi-label thoracic abnormality benchmark comprising over 100,000 images across 14 disease labels [4], and CheXpert extended this paradigm with uncertainty labels and expert comparison protocols [5]. Dedicated COVID-19 chest X-ray datasets further enabled the rapid development of automated screening models during the pandemic [6]. Nevertheless, most existing studies remain confined to binary or limited-class classification rather than a clinically meaningful multi-class distinction encompassing Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia as distinct diagnostic categories.

Prior pneumonia detection studies have demonstrated encouraging results: CheXNet achieved radiologist-level pneumonia detection using DenseNet-based learning [7], and numerous COVID-19 studies applied transfer learning for binary or limited classification tasks [8], [9], [10]. However, direct comparison across studies is difficult due to

heterogeneous datasets, preprocessing pipelines, and evaluation protocols. Furthermore, black-box model predictions without interpretability remain a significant barrier to clinical adoption, as clinicians require evidence that model attention is grounded in genuine pathological findings rather than spurious visual artifacts unrelated to the disease process. Transfer learning is especially valuable in medical imaging, where acquiring large annotated datasets is costly and time-consuming. Pretraining on large natural image datasets enables effective feature adaptation through selective fine-tuning [11], [12]. Explainability is equally essential for clinical trust, as Grad-CAM generates class-discriminative heatmaps identifying image regions most influential to model predictions [13]. Medical XAI research confirms that interpretability enables clinicians to assess whether models focus on clinically relevant anatomical regions [14], and broader XAI frameworks emphasize transparency in high-stakes patient-care domains [15].

This study addresses five key gaps: the predominance of binary classification; limited comparative evaluation of DenseNet121, EfficientNetV2B3, and EfficientNetV2S on a unified dataset and training pipeline; insufficient Grad-CAM integration as a workflow-embedded feature; and the absence of fully deployable clinical decision support systems incorporating secure access, historical prediction management, and automated reporting. The proposed framework performs four-class classification (Normal, COVID-19, Bacterial Pneumonia, Viral Pneumonia), comparatively evaluates the three architectures with transfer learning, integrates Grad-CAM explainability, and delivers an end-to-end web- and mobile-compatible platform with role-based access, prediction history, and automated PDF report generation.

II. LITERATURE REVIEW

Deep learning in computer vision has advanced through a series of landmark architectural developments. The introduction of residual learning resolved the degradation problem in very deep networks through skip connections, enabling stable training of networks exceeding 100 layers and achieving substantial improvements over prior architectures [16]. Prior milestones including deep CNNs with nonlinear activations and dropout regularization, multi-scale inception modules for efficient feature extraction, and very deep networks with small convolutional filters collectively established the representational capacity and training paradigms that underpin modern medical image classification. Visualization techniques further clarified how intermediate convolutional layers respond selectively to image patterns, providing early insight into the learned feature hierarchy of deep networks [17]. DenseNet introduced dense connectivity in which each layer receives feature maps from all preceding layers within a dense block, promoting feature reuse, improved gradient flow throughout the network, and reduced parameter redundancy compared to conventional architectures [18]. This architectural property is particularly advantageous for chest X-ray analysis, where multi-scale feature information is necessary to differentiate between subtle disease-specific patterns such as opacity distributions, peribronchial thickening, airspace consolidation, and ground-glass opacities characteristic of different pneumonia subtypes. DenseNet121 has proven effective in medical imaging tasks due to its capacity for fine-grained pattern capture and parameter efficiency, as evidenced by the influential CheXNet

study that demonstrated densely connected networks achieving radiologist-level performance in pneumonia detection [7].

EfficientNet proposed a principled compound scaling methodology that simultaneously and uniformly scales network depth, width, and input resolution using a fixed set of scaling coefficients [19], overcoming the limitations of ad hoc scaling strategies that independently adjust single dimensions at the expense of others. EfficientNetV2 subsequently refined the original EfficientNet family by introducing Fused-MBConv blocks in earlier network stages, progressive learning with adaptive regularization schedules, and substantially improved training efficiency without sacrificing accuracy [20]. EfficientNetV2B3 is well-suited to deployment-oriented medical imaging applications due to its strong accuracy-to-computational-cost ratio. Prior work on ensemble strategies combining EfficientNetV2-based architectures has demonstrated competitive performance in multiclass chest disease classification, including COVID-19 and other pulmonary conditions [21], supporting the inclusion of EfficientNetV2 variants in this comparative study. ImageNet-based pretraining [22] has become the standard initialization strategy in medical image analysis, supported by both theoretical justification and empirical evidence. Pan and Yang formally defined transfer learning as leveraging knowledge from a source domain to improve learning performance in a related target domain [11], establishing the conceptual foundation for applying natural image pretrained models to medical tasks. Tajbakhsh et al. empirically demonstrated that fine-tuning pretrained CNNs is consistently more effective than training from scratch across a wide range of medical image analysis tasks [12]. Although ImageNet contains natural rather than medical images, early convolutional layers learn generic visual features including edges, textures, and spatial gradients that are transferable across domains. Comprehensive surveys confirm that CNNs have been successfully applied to classification, detection, segmentation, and clinical prognosis in medical imaging, while emphasizing the need for careful methodology to address dataset biases, class imbalance, and spurious correlations [23].

Large chest X-ray benchmarks such as ChestX-ray8 [4] and CheXpert [5] primarily support multi-label thoracic abnormality recognition rather than focused pneumonia subtype classification. Prior studies including CheXNet [7], Apostolopoulos and Mpesiana [8], Ozturk et al. [9], and Narin et al. [10] demonstrated the feasibility of deep learning for pneumonia detection but did not fully address the clinically important four-class distinction among Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia within a unified framework, directly motivating the proposed study. Explainability is a core requirement in medical deep learning. Class activation mapping introduced discriminative region localization in CNNs [17], and Grad-CAM extended this using final convolutional layer gradients to produce precise class-discriminative heatmaps without architectural modifications [13]. Medical XAI surveys stress that interpretability is essential for clinical acceptance and responsible deployment [14], while broader XAI frameworks affirm its indispensability in high-stakes decision-making [15]. Topol argued that AI should augment rather than replace clinicians [1], and Shortliffe and Sepúlveda emphasized the need for interpretable, workflow-integrated outputs for meaningful clinical adoption [3]. Based on the reviewed

literature, five key gaps are identified: (i) most existing studies are binary and do not address multiclass pneumonia subtype classification; (ii) the four-class distinction among Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia remains underexplored in a unified framework; (iii) few studies compare DenseNet121, EfficientNetV2B3, and EfficientNetV2S under the same dataset and training pipeline; (iv) explainability is typically presented as an add-on visualization rather than a workflow-integrated feature with patient record linkage; and (v) deployment aspects including secure authentication, role-based dashboards, historical prediction management, and automated report generation are frequently absent from published systems. The proposed framework directly and systematically addresses all five identified gaps.

III. METHODOLOGY

The proposed methodology encompasses five principal stages: dataset preparation and analysis, preprocessing and augmentation, transfer learning-based model training, comparative evaluation with Grad-CAM explainability integration, and end-to-end clinical system deployment. The overarching objective is to classify chest X-ray images into four clinically meaningful categories Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia while ensuring a reproducible comparison among DenseNet121, EfficientNetV2B3, and EfficientNetV2S. All architectures share one stratified split, a class-weighted loss, and a two-stage protocol; the settings that differed between them are reported in Table II, so observed differences reflect architecture together with configuration.

A. Dataset Description

The dataset used in this study comprises a total of 9,209 chest X-ray images drawn from the public Mendeley curated chest X-ray dataset [24]. The images span four diagnostic categories: COVID-19, Normal, Viral Pneumonia, and Bacterial Pneumonia, as detailed in Table I. The dataset exhibits notable class imbalance, with the Normal and Bacterial Pneumonia classes substantially over-represented relative to the COVID-19 and Viral Pneumonia classes. This imbalance requires explicit mitigation strategies, including class-weighted loss computation and stratified sampling, to prevent the classifier from developing a bias toward majority classes. All images are in JPEG format and vary in acquisition resolution, contrast, and pixel intensity characteristics arising from differences in imaging equipment, patient demographics, and clinical acquisition conditions, further motivating systematic preprocessing and normalization procedures.

TABLE I. NUMBER OF IMAGES PER CLASS IN THE DATASET

Class	Number of Images
COVID-19	1,281
Normal	3,271
Viral Pneumonia	1,656
Bacterial Pneumonia	3,001
Total	9,209

B. Data Preprocessing

Chest X-ray images vary substantially in resolution, contrast, and pixel intensity due to differences in imaging equipment and acquisition protocols, necessitating a systematic preprocessing pipeline applied uniformly across all subsets. Images are resized to 224×224 pixels for DenseNet121 and 300×300 pixels for both EfficientNetV2 models. The single grayscale channel is replicated across three

channels to match the RGB input of ImageNet-pretrained weights. Pixel values are scaled to [0, 1] for DenseNet121, EfficientNetV2B3 normalizes raw pixels internally with ImageNet mean and standard deviation, and for EfficientNetV2S the [0, 1] scaling was followed by the model's built-in [-1, 1] rescaling, compressing inputs into a narrow range. The curators removed duplicate and defective images [24]; no patient-level screening preceded splitting, so same-patient images may span subsets.

C. Data Splitting and Augmentation

The dataset is partitioned into training (6,446 images), validation (1,381), and test (1,382) subsets, a 70/15/15 stratified random split shared by all three models so that class distribution is preserved in every subset. The validation set guides early stopping and checkpointing, and the test set is used only for final evaluation. Augmentation is applied to the training set only: random rotation ($\pm 15^\circ$, or $\pm 20^\circ$ for EfficientNetV2B3), translation and zoom of 10–15%, slight shear, horizontal flipping and, for EfficientNetV2B3, brightness scaling between 0.85 and 1.15. Flipping alters anatomical laterality and is acknowledged as a limitation. Validation and test images undergo only resizing and intensity scaling.

D. Model Architectures

Three architectures representing distinct design philosophies are compared. DenseNet121 [18] consists of four dense blocks of 6, 12, 24, and 16 layers separated by transition layers that apply a 1×1 convolution and 2×2 average pooling. Within a block, every layer receives the concatenated feature maps of all preceding layers and adds 32 new maps through a batch normalization, ReLU, 1×1 bottleneck, and 3×3 convolution sequence. This dense feature reuse retains fine-grained texture alongside abstract context and eases gradient flow, which suits chest X-ray patterns spanning multiple spatial scales.

EfficientNetV2B3 and EfficientNetV2S [20] belong to the compound-scaled EfficientNetV2 family. Their early stages use Fused-MBConv blocks, in which a single 3×3 convolution replaces the expansion and depthwise convolutions, while later stages use MBConv blocks with depthwise convolution and squeeze-and-excitation channel attention. EfficientNetV2S is the deeper and wider variant, enabling a within-family comparison that complements the cross-family analysis. For every backbone the ImageNet classifier is removed and replaced by the same head design: global average pooling, batch normalization, two L2-regularized dense layers with dropout, and a four-class softmax layer. Table II lists the resulting configuration of each model.

E. Transfer Learning Strategy

Transfer learning proceeds in two stages. In the first, all ImageNet-pretrained [22] backbone layers are frozen and only the new head is trained, so the randomly initialized layers converge without large gradient updates disturbing the pretrained weights. In the second, the deepest backbone layers listed in Table II are unfrozen and fine-tuned at a reduced learning rate, so high-level features adapt to chest X-ray characteristics while generic low-level features are preserved. This follows established transfer learning theory [11] and empirical evidence from medical image analysis [12], and the same two-stage protocol is applied to all three architectures.

TABLE II. ARCHITECTURE AND TRAINING CONFIGURATION OF THE THREE MODELS

Setting	DenseNet121	EfficientNetV2B3	EfficientNetV2S
Input size	224×224	300×300	300×300
Input normalization	[0, 1]	ImageNet mean/std	Rescaled twice
Core block	Dense block	Fused-MBConv, MBConv	Fused-MBConv, MBConv
Head dense units	512, 256 (ReLU)	640, 320 (Swish)	512, 256 (ReLU)
Head dropout	0.4, 0.3	0.45, 0.3	0.4, 0.3
Total parameters	7.70 M	14.13 M	21.12 M
Trainable, stage one	0.66 M	1.19 M	0.79 M
Unfrozen, stage two	Final dense block	Last 30% of layers	Blocks 5–6 and top
Batch size	32	24	16
Learning rate, stage one	1×10^{-4}	1×10^{-3}	1×10^{-4}

F. Hyperparameter Configuration

All models are trained with categorical cross-entropy loss and the Adam optimizer, using balanced class weights computed from the training set. Stage one trains the head for up to 10 epochs at the learning rate given in Table II, and stage two fine-tunes for up to 30 epochs at 1×10^{-5} . Early stopping monitors validation accuracy with a patience of 7 epochs (8 for EfficientNetV2B3) and restores the best weights, ReduceLROnPlateau multiplies the learning rate by 0.3 after 4 epochs without validation-loss improvement, and checkpoints are saved on best validation accuracy. Batch size was reduced for the larger models because of memory constraints, and EfficientNetV2B3 used a higher head learning rate; these differences are reported rather than equalized. Each model was trained once with a fixed seed, so run-to-run variance is not estimated.

G. Evaluation Metrics

Performance is evaluated using accuracy, macro-averaged precision, recall, and F1-score, micro-averaged one-vs-rest AUC, per-class recall, and confusion matrix analysis. Accuracy is reported with a 95% Wilson confidence interval, because a single test set of 1,382 images limits how finely models can be distinguished. Overall accuracy is insufficient under class imbalance, and class-wise recall is particularly critical in medical imaging, because missed pneumonia or COVID-19 cases can delay treatment. Final model selection

balances overall accuracy, class-wise performance, explanation quality, and deployment feasibility.

H. Grad-CAM Explainability

Grad-CAM [13] is computed on the final convolutional block of the deployed model, a $7 \times 7 \times 1024$ map for DenseNet121. Gradients of the predicted class score with respect to these feature maps are globally average-pooled to give channel weights, which are combined with the activation maps and passed through a ReLU to retain positively contributing regions; the result is normalized by its maximum. The class score is taken from the pre-softmax logits, because the saturated softmax of a confident prediction yields vanishing gradients. The logits are reconstructed by applying the final dense layer's weights without its softmax, so the deployed classifier is unchanged. The heatmap is bilinearly upsampled to the image resolution and restricted to the lung fields segmented by a pretrained U-Net [25], so no heat is displayed outside the lungs (Fig. 1). The U-Net receives a 256×256 histogram-equalized, standardized image; its probability map is thresholded at 0.5, the two largest regions centred above the lowest fifth of the image are kept, holes are filled, and the edge is softened without enlarging the field. If no plausible lung field is found, the unmasked map is shown. The fraction of attention inside the lungs is computed on the unmasked map and reported, the masked map is rescaled to its own peak, and a caution is shown when the fraction falls below one half. Grad-CAM is not intended to replace expert radiological interpretation but to improve prediction transparency and serve as visual decision-support evidence.

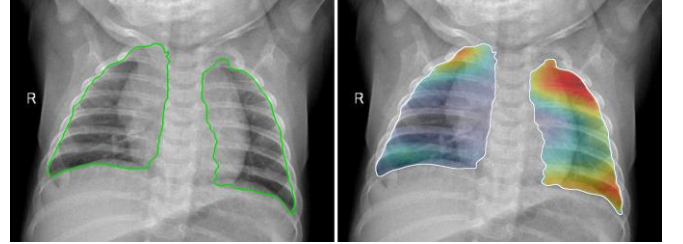


Fig. 1. Grad-CAM from the deployed system for a radiograph classified as Normal (99.9%): U-Net lung fields (left) and lung-restricted heatmap (right).

I. System Deployment Architecture

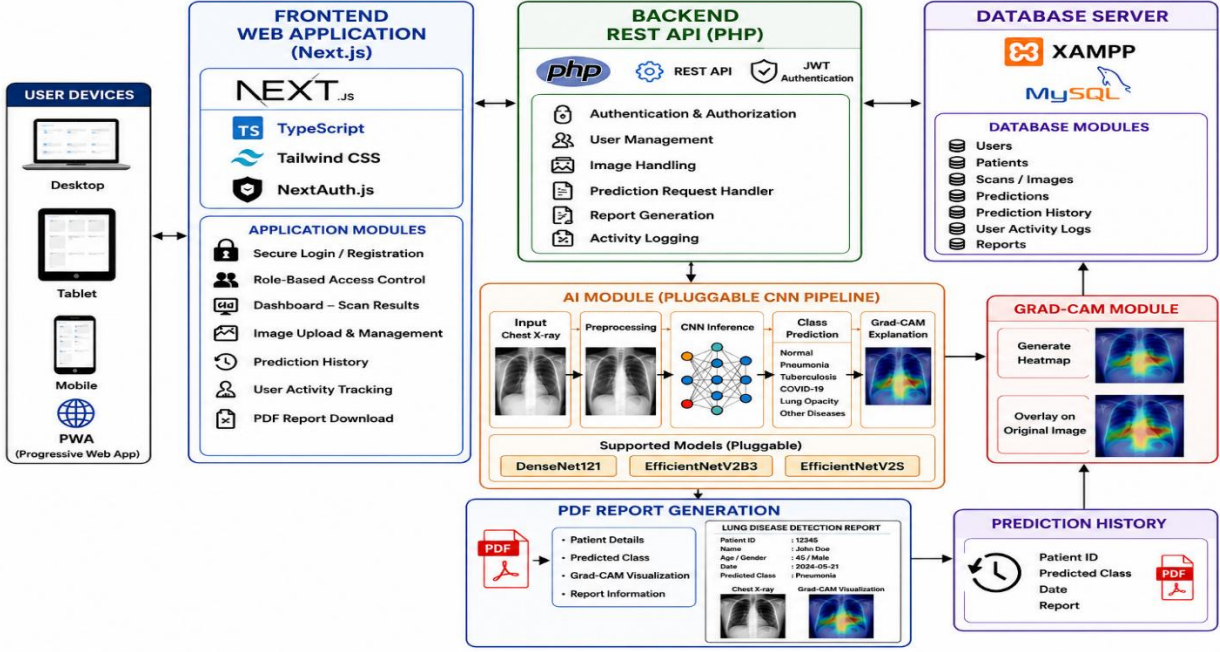


Fig. 2. End-to-End System Deployment Architecture of the Web-Based Lung Disease Detection Platform.

The system is deployed as a web-based, mobile-friendly lung disease detection platform for desktops, tablets, and mobile devices. The frontend uses Next.js, TypeScript, Tailwind CSS, and NextAuth.js, while the backend uses PHP REST APIs, JWT authentication, and MySQL via XAMPP, providing secure login, role-based access control (patient, doctor, researcher, administrator), an administrative panel for managing users, scans, activity, and statistics, and account self-management (profile editing, password change and recovery, and deletion). Dashboards present scan results, confidence visualization, image management, prediction history, and CSV export. Inference is decoupled into a dedicated Python (FastAPI) microservice that loads the trained Keras models and runs a pluggable CNN pipeline supporting DenseNet121, EfficientNetV2B3, or EfficientNetV2S, performing preprocessing, prediction, and confidence scoring. A post-hoc out-of-distribution safeguard, combining Mahalanobis-distance, energy, and maximum-softmax signals, withholds the diagnosis for invalid inputs, recording the verdict, score, and reason. Results are stored, and automated PDF reports in English, Sinhala, and Tamil contain patient details, predicted class, confidence, and timestamp. An assistance chatbot, light and dark modes, and informational pages complete a secure end-to-end clinical decision support system. Fig. 2 illustrates the end-to-end system deployment architecture of the proposed web-based lung disease detection platform, encompassing the AI prediction pipeline, out-of-distribution safeguard, secure user authentication, and automated multilingual PDF report generation.

IV. RESULTS AND COMPARATIVE ANALYSIS

Table III reports test-set performance on the 1,382 held-out images, never used for training or checkpointing. DenseNet121 achieved the highest accuracy of 85.31%, followed by EfficientNetV2B3 with 84.30% and EfficientNetV2S with 80.54%. The 1.01-point gap between the first two models corresponds to 14 images from a single training run, and their 95% confidence intervals overlap, so the two are not distinguishable. The lower EfficientNetV2S

result should not be read as an architectural finding, because its inputs were rescaled twice into a narrow range its pretrained weights do not expect, and its best validation accuracy came in the 29th of 30 fine-tuning epochs. EfficientNetV2S is also the largest of the three models, so it offers no deployment advantage here.

TABLE III. TEST-SET PERFORMANCE OF THE THREE MODELS (N = 1,382; MACRO-AVERAGED PRECISION, RECALL, AND F1-SCORE)

Model	Accuracy % (95% CI)	Prec.	Recall	F1	AUC
DenseNet121	85.31 (83.3–87.1)	0.84	0.85	0.84	0.977
EfficientNetV2B3	84.30 (82.3–86.1)	0.83	0.83	0.83	0.970
EfficientNetV2S	80.54 (78.4–82.5)	0.80	0.81	0.80	0.960

Per-class results show where the errors lie. All three models recognized COVID-19 and Normal radiographs with recall between 0.97 and 0.99, whereas bacterial pneumonia recall ranged from 0.61 to 0.79 and viral pneumonia recall from 0.55 to 0.69. For DenseNet121, 155 of its 203 test errors (76%) were confusions between bacterial and viral pneumonia, while only 3 of 192 COVID-19 and 5 of 491 Normal images were misclassified. The system therefore separates abnormal from normal radiographs reliably, but its bacterial–viral distinction, the one most relevant to antibiotic decisions, is not yet accurate enough to guide treatment.

TABLE IV. PROPOSED FUNCTIONAL COMPONENTS OF THE CLINICAL DECISION SUPPORT SYSTEM

Component	Function
Image Upload	Accepts chest X-ray images through the web interface
CNN Prediction Module	Classifies images into Normal, COVID-19, Bacterial Pneumonia, or Viral Pneumonia
Confidence Visualization	Displays class probability and prediction confidence
Grad-CAM Explanation	Highlights image regions influencing the prediction
Prediction History	Stores previous scan results for patient management
PDF Report Generator	Produces structured medical reports
Web Dashboard	Supports user, scan, and result management
Mobile-Friendly PWA	Enables app-like access across devices
Secure Authentication	Uses JWT authentication and role-based access control

Table IV lists the functional components of the proposed system. The integration of Grad-CAM allows users to inspect whether the model focuses on relevant lung regions during prediction, which matters when two models achieve similar accuracy: a model with slightly lower accuracy but more clinically meaningful heatmaps may be more useful than one with marginally higher accuracy but diffuse attention.

Grad-CAM outcomes were measured with the deployed DenseNet121 pipeline, whose final feature map is $7 \times 7 \times 1024$ for a 224×224 input, so each heatmap cell spans about 32×32 pixels and the maps indicate regions rather than lesion boundaries. On the Fig. 1 radiograph, the gradient of the softmax score was 467 times smaller than that of the logit, and the softmax-based map peaked at 0.0004 against 0.22, confirming why logits are used. Only 19% of the unmasked attention fell inside the segmented lungs, which covered 23% of the image, so the platform raised a caution. Agreement with occlusion sensitivity was weak (Pearson $r = 0.28$ over the image, 0.30 within the lungs) and unchanged by masking, so masking improves presentation, not faithfulness. On 12 ChestX-ray8 [4] radiographs and this scan, the U-Net delineated the lungs in all but one fully opacified case, where the unmasked map is shown, and found no lungs in a random-noise or a logo image, both flagged as out of distribution. No map was compared with expert radiologist annotations.

TABLE V. COMPARISON WITH PREVIOUSLY REPORTED CHEST X-RAY CLASSIFICATION METHODS

Study	Model	Classes	Reported
Rajpurkar et al. [7]	DenseNet121	2	F1 0.435
Apostolopoulos and Mpesiana [8]	VGG19	3	Acc. 93.48%
Ozturk et al. [9]	DarkCovidNet	3	Acc. 87.02%
Narin et al. [10]	ResNet50	2	Acc. 96.1%
Chowdhury et al. [6]	DenseNet201	3	Acc. 97.9%
This work	DenseNet121	4	Acc. 85.31%

Table V compares these results with earlier methods for similar prediction, as reported by their authors on different datasets and class definitions, so the comparison is indicative rather than head-to-head. The studies reporting above 93% address binary or three-class tasks that do not separate bacterial from viral pneumonia; the closest three-class result, 87.02% [9], is of the same order as the 85.31% obtained here on the harder four-class task. Unlike the listed studies, the proposed framework quantifies Grad-CAM attention inside the lungs and checks input validity before reporting a diagnosis.

V. DISCUSSION

The findings indicate that explainable transfer learning is a workable approach for four-class pneumonia classification, with three qualifications. First, accuracy differences between architectures are small relative to their uncertainty and partly reflect configuration, so the evidence does not establish that one design is superior; DenseNet121 was deployed because it led on every metric in Table III. Second, the system separates abnormal from normal radiographs reliably but not bacterial from viral pneumonia, so it suits screening and triage support rather than aetiological diagnosis. Third, Grad-CAM shows where the model attends, not why, and may place much attention outside the lungs even on confident predictions; disclosing the in-lung share makes this visible to clinicians. All results come from one retrospective public dataset, so validation across hospitals, devices, and demographic groups, together with radiologist review of explanations, is required before clinical use. The system is therefore positioned as a decision-assistance tool that supports rather than replaces medical judgment.

VI. CONCLUSION

This paper presented an explainable deep learning framework for four-class classification of chest X-ray images into Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia. Under a shared split and two-stage transfer learning protocol, DenseNet121 achieved 85.31% accuracy, EfficientNetV2B3 84.30%, and EfficientNetV2S 80.54%, with most errors arising between the two pneumonia subtypes. Beyond model comparison, the work contributes an end-to-end clinical decision support framework that integrates prediction, logit-based and lung-restricted Grad-CAM, historical prediction management, mobile-friendly access, secure authentication, and automated PDF report generation.

The proposed system is best viewed as an assistive tool that provides rapid predictions and interpretable visual evidence while clinical authority remains with medical professionals. This study has several limitations. Evaluation used a single public dataset and one training run per model, without external validation; training settings were not fully equalized across architectures; horizontal flipping was used despite anatomical laterality; and Grad-CAM maps were not validated against radiologist annotations. Future work will address these through external validation on diverse chest X-ray datasets, repeated and fully controlled training, radiologist assessment of explanations, integration of multimodal clinical data, and model optimization for resource-constrained settings.

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